When the Treatment Talks Back: Information Limits of Observational Logs from Adaptive Generative Political Communication*

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August 31, 2026

Abstract

Conversational artificial intelligence now conducts political persuasion at scale, and the systems that do so adapt: they choose what to say at each turn as a function of what the person has said so far. Regulating or replicating such a system requires knowing which of its behaviors move beliefs, and the natural evidence base is the deployment log. We show that this evidence base is limited in a way that no estimator can repair, and we characterize the limit.

We define the treatment delivered by an adaptive generative system as a triple: a strategy policy, a generation kernel, and an information environment. For the turn-level excursion effect of such a system we derive the efficient influence function, which yields a semiparametric efficiency bound growing exponentially in a policy decisiveness index. The index is a functional of the behavior policy alone and is identified from logs. Softmax temperature is one special case of it, and the classical average-treatment-effect bound is the degenerate case of our bound. The consequence is not merely a large constant. Along policy sequences whose decisiveness grows with sample size, a two-point argument bounds the minimax risk away from zero: no estimator recovers turn-level effects at any rate. The regime in which estimates appear stable while remaining wrong is exactly the regime in which policies with different effects are statistically indistinguishable.

The information that survives is located rather than merely diminished. As a policy sharpens, its overlap mass concentrates on the decision boundary the policy has learned. The estimand that remains is a boundary effect in the regression-discontinuity sense, defined at a threshold chosen by the system rather than the analyst, and therefore describing the minority of histories about which the system is undecided. We prove this in one dimension and, by a coarea argument, on a decision manifold in $\mathbb{R}^d$. Re-query experiments on three open-weights models suggest deployed policies are sharp enough for these limits to bind, subject to measurement caveats we report in full. The implication for research design is direct. Turn-level causal knowledge about conversational AI must be randomized into the conversation rather than recovered from it, and we specify the protocol that does so.

*Replication code and all numerical results: `github.com/kunhailP/when-treatment-talks-back`. Every number reported here is produced by executed code in that repository, and `scripts/validate_repo.sh` checks the manuscript's figures against the result files they come from.

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1 Introduction

Conversational artificial intelligence persuades. In randomized trials, GPT-4 shifted debate opponents' positions more often than human debaters did (Salvi et al., 2025), and conversations with frontier models moved policy attitudes with effects that persisted for weeks (Hackenburg et al., 2025). Against the benchmark of political persuasion research, where meta-analytic average effects fall well below one percentage point (Kalla and Broockman, 2018), these are substantial.

Establishing that a technology persuades is the first question, not the last. The objects of political regulation and scientific replication are narrower than the technology. They are behaviors: the system asserted rather than asked, or supplied five facts rather than one, at this point in this conversation. Which of those behaviors carries the effect determines what a platform should be required to disclose, what a countermeasure should target, and what a replication would have to hold fixed. The strongest single lever identified by Hackenburg et al. (2025), a reward model selecting among candidate responses turn by turn, is precisely a turn-level adaptive policy. Its logs are what an analyst inherits.

The natural response is to treat those logs as observational data on a time-varying treatment and to apply the corresponding identification machinery. This paper explains why that response fails. The failure is not the familiar one of unmeasured confounding, which a richer log might mitigate. It is specific to adaptive systems. A policy that has learned which action is better at a given history stops taking the other one. The counterfactual contrast the analyst requires is exactly the contrast the system has learned not to produce. Competence at the task and informativeness about the task stand in tension, and that tension can be quantified.

Three results follow. Where a deployed policy is history-deterministic, the turn-level effect is not identified at all. This is a support failure, standard since Robins (1986). What is new is that preference optimization widens the score gaps that produce it and low-temperature deployment sharpens them further, so the triggering condition is systematic rather than accidental. Where positivity survives, it survives without much use. The efficiency bound for the turn-level effect grows exponentially in how decisive the policy is, and along sequences of increasingly decisive policies the minimax risk does not converge to zero at any sample size. Finally, the information that does survive is not a weaker estimate of the population effect but a different estimand. It lives on the policy's decision boundary, where the policy is undecided, which in a confident deployment is a shrinking minority of interactions.

That last result is where a negative finding becomes usable. A boundary estimand at a learned threshold is a regression-discontinuity quantity, and it inherits the interpretive apparatus of that design, including the understanding that it does not describe the population the deployment actually served. An analyst who reports an effect estimated from adaptive logs is, in the favorable case, reporting this quantity, and should say so.

Because the obstruction is created by the deployed policy before any analysis begins, no estimator removes it. Only a different design does. Section 6 develops the generative micro-randomized trial: randomize the action within the conversation, lock the evidence available to the generator, and treat the generate-and-check procedure as the treatment kernel, so that the identified quantity is assignment to a generative protocol rather than the effect of a realized sentence. The experiment itself is a companion paper. What belongs here is the argument that it is necessary.

Relation to existing work. Three literatures are adjacent. Imai and Nakamura (2026) and Nakamura and Imai (2026) identify effects of features of generated text, including sequences of features, using internal representations of the generating model. Overlap in that setting is established rather than assumed, under a separability condition, and those authors warn that naive adjustment on learned representations destroys it. Our object differs in where overlap is lost. There it is a property of the representation the analyst conditions on. Here it is destroyed by the deployed policy before analysis begins. Second, the estimation-side face of the problem appears as off-policy evaluation under deficient support (Sachdeva et al., 2020; Dudík et al., 2011; Thomas and Brunskill, 2016; Jiang and Li, 2016; Kallus and Uehara, 2020). That literature bounds the behavior of particular estimators. Our statements concern the efficiency bound and the minimax risk, which closes the response that a better estimator would suffice. Its remedies, such as truncation and support restriction, change the estimand without characterizing the result; Section 4 supplies that characterization. Third, the econometrics of limited overlap (Khan and Tamer, 2010; D'Amour et al., 2021; Li et al., 2018) is the closest antecedent for the shape of these results. Our contribution is to index the failure by a policy functional that a deployment itself determines, and to identify the estimand that survives. Micro-randomized trials (Qian et al., 2022; Boruvka et al., 2018) supply the excursion estimand we adopt, and the theory of stochastic interventions (Díaz Muñoz and van der Laan, 2012; Kennedy, 2019; Young et al., 2014) supplies its formal object. What conversation adds is a generation kernel between assignment and realization, and the fidelity problem that follows.

2 Generative Treatments

Individuals $i = 1, \dots, n$ converse over turns $t = 1, \dots, T$. At turn t the system draws an action from its policy, $A_{it} \sim \pi_t(\cdot \mid H_{it})$, where H_{it} is the interaction history, and realizes an utterance from its generation kernel, $M_{it} \sim G_\theta(\cdot \mid A_{it}, H_{it}, E_j)$, where E_j is the information environment (for us: a locked evidence bank on issue j). The user responds R_{it}; outcomes Y_i include belief accuracy, calibration, reactance, and one-week retention.

The action space is general. $A_t \in \mathcal{A}$ may be a discrete strategy label, a continuous generation parameter, or a factorial cell. As a running example, and as the implementation object of the companion experimental paper, we use a 2×2: form $F \in \{\text{assertive}, \text{interrogative}\} \times$ dose $D \in \{\text{high: 4–5 facts}, \text{low: 1–2 facts}\}$.

Treatment is a stochastic intervention. Potential outcomes are defined under regimes, $Y_i^{\pi,G}$, with do-notation read as a kernel intervention: intervene on the label, realize from G (Díaz Muñoz and van der Laan, 2012; Young et al., 2014). Three estimand layers:

- **Deployment effect** $\Delta^{\text{dep}} = \mathbb{E}[Y^{\pi_1,G_1}] - \mathbb{E}[Y^{\pi_0,G_0}]$: what existing RCTs identify.
- **Policy effect** $\Delta^{\text{pol}}(G) = \mathbb{E}[Y^{\pi_1,G}] - \mathbb{E}[Y^{\pi_0,G}]$: holding the generator fixed.
- **Turn-level excursion effects** (primary): contrasts of one turn's action against a pre-specified reference policy ρ (uniform over $\mathcal{A}$, all t, all h), with subsequent turns following ρ. Proximal excursions (immediate reactance, belief update) are standard CEE/WCLS targets (Boruvka et al., 2018); distal excursions (final accuracy, retention) connect to distal causal excursion effect estimation (Qian, 2025). Using ordinary WCLS for terminal outcomes is a known attack point, which this distinction avoids.

Ill-defined treatment in logs. An observational label $\tilde{S}_t = s$ corresponds to the realized kernel $M_t \sim G(\cdot \mid s, H_t)$, a *different distribution at every history.* Label-pooled contrasts do not correspond to any well-defined treatment pair. Our regime definition does not "solve" this nonidentification; it replaces the question with a well-posed one, and the well-posed β is identified only under assignment randomization (Section 6).

3 The Information Limit

The results below are stated for a general policy through the decisiveness index of Definition 1. It helps to keep one parameterization in view: a policy choosing softmax in a history-dependent score gap at effective temperature τ, so that $p_\tau(h) = \sigma(\Delta(h)/\tau)$ in the two-action case. Deployment practice pushes τ down; preference optimization pushes $|\Delta|$ up. Nothing in the theorems requires this form.

3.1 Support failure

Proposition 1 (Support failure)**.** *If $\pi_t^{\text{dep}}(a \mid h) = 0$ on a positive-probability set of histories, turn-level excursion effects are nonparametrically unidentified.*

The counterexample pair requires no latent confounder, the failure is support, not confounding. Take $T = 1$, $H \in \{h_0, h_1\}$, $\pi^{\text{dep}}(1 \mid h_0) = 1/2$, $\pi^{\text{dep}}(1 \mid h_1) = 0$, $Y(a) \mid H = h \sim N(\mu_a(h), 1)$, and let two DGPs differ only in $\mu_1(h_1) = c_1$ versus c_2. The observed-data law factorizes as $P(H)\,\pi^{\text{dep}}(A \mid H)\,N(y; \mu_A(H), 1)$, to which the $(1, h_1)$ stratum contributes nothing; the two DGPs are observationally identical yet their uniform-reference excursion effects differ by $P(h_1)(c_2 - c_1)$. The multi-turn version replaces H with $(X, R_{1:t-1})$ and recovers the time-varying-confounding form. The structure is Robins (1986); the LLM-specific claim is frequency and reinforcement, not inevitability.

3.2 Exponential collapse of the information bound

With $\tau > 0$, positivity holds formally, so the right response to Proposition 1 alone would be "use IPW." Proposition 2 closes that escape at the level of the efficiency bound.

Proposition 2 (Result A)**.** *Let $D(h) = -\log \min_{a \in \mathcal{A}^*} \pi(a \mid h)$ be the policy's decisiveness at h (Definition 1), where $\mathcal{A}^*$ is the set of actions entering the contrast. Assume (A1′) $\mathbb{P}\{D(H) \geq d\} \geq c > 0$; (A2′) conditional outcome variances on that set are bounded below by $\underline{\sigma}^2$; and (A3) the model class is nonparametric in the outcome regression, no parametric extrapolation into unobserved strata. Then the semiparametric efficiency bound for the turn-level excursion effect satisfies*

$$V \;\geq\; \underline{\sigma}^2\,\rho_{\min}^2\,\mathbb{E}\big[e^{D(H)}\big] \;\geq\; \underline{\sigma}^2\,\rho_{\min}^2\,c\,e^d,$$

with $\rho_{\min}$ the smallest reference probability on $\mathcal{A}^*$.

We derive this from the efficient influence function of the excursion estimand itself (Proposition 6 and Appendix A) rather than importing a bound stated for the average treatment effect. The ATE bound of Hahn (1998) is recovered as the two-action, degenerate-reference special case. Two consequences are worth isolating. First, the governing quantity is a property of the behaviour policy alone and is identified from logs; a softmax at temperature τ gives $D = |\Delta|/\tau$ as one special case, but the statement covers truncated and constrained decoding, hard filters, and learned arg max rules, none of which are softmax. Second, in a factorial design the relevant probabilities are the *cell* probabilities, not the marginal for the factor being contrasted: because the reference regime draws the other factor from ρ, a design in which the form-marginal stays at $1/2$ while cells become rare still has an exploding bound.

By the convolution theorem, *no regular estimator in the class* attains lower asymptotic variance: switching from IPW to AIPW recovers no counterfactual information that the sample does not contain. Parametric extrapolation escapes the bound, but that is buying identification with an assumption, and should be said in exactly those words. Effective sample size is arm-specific Kish: $n_{\text{eff}}(a) = n/\mathbb{E}[1/\pi_\tau(a \mid H)]$. The slogan "halving τ squares the required n" holds within fixed-gap strata; with heterogeneous gaps the max-gap stratum dominates the rate.

Remark 1 (Single-turn bounds are a fortiori multi-turn bounds). Our results are stated for one turn and two actions, while the motivating setting is a T-turn conversation. The gap runs in the harmless direction. The turn-t excursion estimand conditions on the observed history H_t and contrasts actions at that turn, with subsequent turns following ρ. The observed-data model factorizes turn by turn; the turn-t factor is exactly the single-turn problem with propensity $\pi_\tau(\cdot \mid H_t)$, and identifying β_t additionally requires bridging every later turn from π^{dep} to ρ. Discarding the information cost of those later-turn bridges can only *understate* the difficulty; hence any lower bound for the single-turn problem at the (Δ, τ) prevailing at turn t is also a lower bound for the conversational problem. (K-action policies reduce to the two-action case via the two arms of the contrast.)

Proposition 3 (A-T: multi-turn compounding). *Consider the T-turn policy value $V(\rho)$ of the uniform reference ρ over K actions, observed under π_τ. Start from the semiparametric efficiency bound for finite-horizon off-policy evaluation with history-dependent policies (Kallus and Uehara, 2020; Jiang and Li, 2016):*

$$V_{\text{eff}} = \mathrm{Var}(V_1(H_1)) + \sum_{t=1}^{T} \mathbb{E}_\pi\Big[\Big(\textstyle\prod_{s\le t} w_s\Big)^2 \mathrm{Var}(V_{t+1} \mid H_t, A_t)\Big], \qquad w_s = \tfrac{\rho(A_s|H_s)}{\pi_\tau(A_s|H_s)}.$$

Assume (M1) uniform recurrence of the high-gap set*: $\mathbb{P}(H_{s+1} \in \mathcal{H}_\delta \mid h, a) \ge p_\delta > 0$ for every $h \in \mathcal{H}_\delta$ and every action a; and (M2) a per-turn conditional variance floor $\underline{\sigma}^2$ on $\mathcal{H}_\delta$. Then*

$$V_{\text{eff}} \;\ge\; \underline{\sigma}^2 \cdot \mathbb{P}(H_1 \in \mathcal{H}_\delta) \cdot p_\delta^{T-1} \cdot \left[\frac{e^{\delta/\tau}}{K^2}\right]^T,$$

which compounds geometrically in T whenever $\tau < \delta/\log(K^2/p_\delta)$.

Proof sketch. All terms of V_{eff} are nonnegative; keep the terminal one. On $\mathcal{H}_\delta$, $\mathbb{E}_\pi[w_t^2 \mid H_t = h] = K^{-2}\sum_a 1/\pi_\tau(a \mid h) \ge K^{-2}e^{\delta/\tau}$ since the rarest action has probability at most $e^{-\delta/\tau}$. Induct backward from $t = T$: by the tower property and (M1), whose action-uniformity is what lets the bound pass through each turn's weight, each turn contributes a factor $p_\delta\, e^{\delta/\tau}/K^2$, and turn 1 contributes $\mathbb{P}(H_1 \in \mathcal{H}_\delta)\, e^{\delta/\tau}/K^2$. We quote rather than re-derive the finite-horizon efficiency bound; the form above is the discrete-horizon Cramér–Rao bound of Jiang and Li (2016), which Kallus and Uehara (2020) extend to history-dependent policies. Our contribution is the insertion and the induction. ☐

Remark 2 (Scope of Proposition 3). Two limitations should be read with this proposition. It bounds the *policy value* $V(\rho)$, which we treat as a secondary estimand (Section 2); the corresponding statement for the primary turn-level excursion effect requires redoing the induction with the continuation weights of Appendix A and we do not claim it here. And (M1) requires the history to remain in $\mathcal{H}_\delta$ under *every* action, i.e. that the treatment does not move the history state much, close to the opposite of the premise in this paper's title. The per-turn factor is $p_\delta e^{\delta/\tau}/K^2$, and p_δ is not estimated anywhere in this paper, so multi-turn compounding should be read as conditional on an unverified quantity rather than as an empirical claim.

Substantively, (M1) says decisiveness persists regardless of what the system does, as with strongly reactant partisan users. The K^2 factor reflects the conservativeness of the uniform reference and is not optimized; the excursion version (turn t, reference thereafter) follows with exponent $T - t + 1$. The honest reading: more turns make the information limit exponentially worse, now as a proposition rather than a heuristic.

Proposition 4 (Minimax collapse)**.** *Let $M(\pi) := \sup_{d>0} \mathbb{P}\{D(H) \geq d\}\, e^d$. Consider the model class in which π is known and fixed, $Y \mid A = a, H = h \sim N(\mu_a(h), \sigma^2)$, and $|\mu| \leq B$. Then for any estimator $\hat\beta$,*

$$\inf_{\hat\beta} \sup_{P} \ \mathbb{E}_P\big|\hat\beta - \beta(P)\big| \ \geq \ \tfrac{1}{4} \min\Big\{ B\rho_{\min},\ \rho_{\min}\, \sigma \sqrt{M(\pi)/n} \,\Big\}.$$

In particular, along a sequence of policies with $M(\pi_n)/n \to \infty$ the minimax risk is bounded away from zero: no estimator recovers β at any rate.

This is the statement that Proposition 2 alone does not support. Proposition 2 bounds the information constant at a fixed policy, where β remains regularly $\sqrt{n}$-estimable; Proposition 4 says that once decisiveness grows fast enough relative to sample size, estimability itself fails. The proof (Appendix A) is a two-point argument in which the two laws differ only in the outcome regression at the rarest action on the high-decisiveness stratum, so that they are separated in the estimand but nearly indistinguishable in the data. For the main design $H \sim U(-1, 1)$, $M = \tau e^{(1-\tau)/\tau}$ in closed form, i.e. $\log M = 1/\tau - 1 - \log(1/\tau)$.

The deceptive regime is this indistinguishability. In the main simulation design, two laws whose true ATEs differ by 0.198, differing only in β on the rarely-treated region, are separated by 0.16 standard deviations of the Hájek estimator at $\tau = 0.08$, $n = 2{,}000$, and still only 0.40 at $n = 32{,}000$, a sixteenfold increase in sample size. At $\tau = 0.22$ the same 0.200 gap is recovered as 0.224 at 1.69 standard deviations. What the phase diagram labels "biased but stable" is not a description of estimator behaviour but an instance of two-point indistinguishability.

3.3 What the assumptions require

The results above rest on conditions that are worth reading substantively, because they determine which deployments the limits describe.

Positive mass at high decisiveness (A1′) asks that a non-negligible share of histories be ones on which the policy is close to committed. This is the condition that makes the results about deployment rather than about a pathological corner. It is testable from logs whenever the action probabilities are recorded, and Appendix E estimates a proxy for it.

A variance floor (A2′) asks that outcomes not be deterministic given history and action on that region. Without it a rare action could be uninformative and yet cost nothing to miss. In belief-change outcomes with measurement error the condition is mild.

A nonparametric outcome model (A3) is the substantive commitment. If the analyst is willing to assume a parametric outcome regression that extrapolates into histories where an action was never taken, the bound does not apply, because the missing counterfactual is supplied by the functional form rather than by the data. That is a coherent position. It purchases identification with an assumption that the data cannot check, which is the trade the bound is designed to make visible.

Regularity for localization (B1)–(B4) and (C1)–(C5) require the score to be continuously differentiable with a non-degenerate boundary: zero must be a regular value of Δ, and the boundary must carry positive density. The first fails if the policy's indifference set has a flat or self-intersecting geometry, in which case the limiting measure is not a surface measure. The second is what keeps the statement non-vacuous, and it fails if the policy is indifferent only on histories that never occur, which is the case of a deployment with no equipoise at all.

Uniform recurrence (M1), used only for the multi-turn statement, asks that a history in the high-decisiveness region stay there whatever the system does. Remark 2 discusses why this sits uneasily with the premise of an adaptive system that changes the state it is acting on, and why we treat the multi-turn result as conditional.

Two assumptions are conspicuously *not* required. Nothing assumes the policy is a softmax; that parameterization is an illustration, and Definition 1 is stated for any policy with positive action probabilities. And nothing assumes the analyst knows the policy: the results are bounds on what any analysis could achieve, so knowing π exactly, as in our oracle simulations, does not evade them.

4 Where the Surviving Information Lives

Proposition 5 (Result B, one-dimensional)**.** *Let $\omega_\tau(H) = p_\tau(1-p_\tau)$ and $\beta_\tau^{\mathrm{ov}} = \mathbb{E}[\omega_\tau\{Y(1) - Y(0)\}]/\mathbb{E}[\omega_\tau]$. Assume:*

- *(B1) H has a density f, continuous in a neighbourhood of each root;*
- *(B2) $\Delta \in C^1$ and the root set $\mathcal{R} = \{h : \Delta(h) = 0\}$ is* finite, *with every root simple ($\Delta'(r) \neq 0$);*
- *(B3) $c(h) = \mathbb{E}[Y(1) - Y(0) \mid H = h]$ is bounded, and continuous near each root;*
- *(B4) $f(r) > 0$ for at least one $r \in \mathcal{R}$.*

together with the tail-separation condition (B2′): $\inf_{\mathrm{dist}(h,\mathcal{R})\geq\epsilon} |\Delta(h)| > 0$ for every $\epsilon > 0$. Then, as $\tau \downarrow 0$, the normalized overlap measure $\omega_\tau f / \int \omega_\tau f$ converges weakly to $\sum_j w_j \delta_{r_j}$ with weights $w_j \propto f(r_j)/|\Delta'(r_j)|$. In the special case of a unique *root at 0 the limit is a single point mass and*

$$\mathbb{E}[\omega_\tau] = \tau\,\frac{f(0)}{|\Delta'(0)|}\,(1 + o(1)), \qquad \beta_\tau^{\mathrm{ov}} \to \mathbb{E}[Y(1) - Y(0) \mid H = 0].$$

Corollary 1 (Decision-manifold localization, $H \in \mathbb{R}^d$)**.** *Let $H \in \mathbb{R}^d$ with density f, let $\mathcal{B} = \{h : \Delta(h) = 0\}$, and assume: (C1) $\Delta \in C^1$ and f is bounded and continuous near $\mathcal{B}$; (C2) 0 is a* regular value *of Δ, i.e. $\nabla\Delta(h) \neq 0$ on $\mathcal{B}$, the d-dimensional analogue of the simple-root condition (B2); (C3)–(C4) the tail-separation condition of Proposition 5 holds with* $\mathrm{dist}(\cdot, \mathcal{B})$*; and (C5) $0 < \int_{\mathcal{B}} f/\|\nabla\Delta\|\, d\mathcal{H}^{d-1} < \infty$. Then as $\tau \downarrow 0$,*

$$\mathbb{E}[\omega_\tau] \;=\; \tau \int_{\mathcal{B}} \frac{f}{\|\nabla\Delta\|}\, d\mathcal{H}^{d-1}\,(1 + o(1)),$$

and the normalized overlap measure converges weakly to the measure on $\mathcal{B}$ with $\mathcal{H}^{d-1}$-density proportional to $f(h)/\|\nabla\Delta(h)\|$. Consequently $\beta_\tau^{\mathrm{ov}} \to \int_{\mathcal{B}} c\,f/\|\nabla\Delta\| / \int_{\mathcal{B}} f/\|\nabla\Delta\|$. Proposition 5 is the case $d = 1$, where $\mathcal{H}^0$ is counting measure and $\|\nabla\Delta\| = |\Delta'|$.

Remark 3 (What β^{ov} is). The limit in Corollary 1 is a boundary estimand in the regression-discontinuity sense: in one dimension it is exactly $\mathbb{E}[Y(1) - Y(0) \mid \Delta(H) = 0]$, the conditional effect at the assignment threshold (Hahn et al., 2001), and in d dimensions it is the $f/\|\nabla\Delta\|$-weighted average effect over a boundary, the object studied in multi-dimensional and geographic RD (Keele and Titiunik, 2015). The substantive reading is sharper than an overlap-weighting analogy: logs from a sharply adaptive system behave less like an observational study than like a natural experiment at a threshold, but the threshold is one the *policy* learned, not one the analyst chose. What is stably estimable therefore describes the histories on which the policy is *undecided*, which in a decisive deployment is precisely the minority of traffic.

Remark 4 (The surviving information also vanishes). Proposition 5 says *where* overlap mass concentrates, not how much of it there is. Since $\mathbb{E}[\omega_\tau] = O(\tau)$, the effective sample available for β_τ^{ov} is itself $O(n\tau)$ and degenerates in the same limit. Result B should therefore not be read as saying that boundary information is *stable*: what survives longest is boundary information, but it too disappears as the policy sharpens. Assumption (B4) is exactly what prevents the leading term from collapsing to $0 = o(\tau)$; without it the statement is vacuous.

Proof idea. $\omega_\tau = K(\Delta(h)/\tau)$ exactly, with $K(u) = 1/[4\cosh^2(u/2)]$ and $\int K = 1$: an approximate-identity kernel. The tail-separation condition kills contributions away from the roots at rate $e^{-c(\epsilon)/\tau}$; near a simple root, the change of variables $v = \Delta(h)$, $v = \tau u$ and dominated convergence give $\int g\,\omega_\tau f\,dh = \tau\, g(0)f(0)/|\Delta'(0)| + o(\tau)$ for bounded continuous g. Ratios deliver the three claims. With multiple roots, mass distributes proportionally to $f(r_j)/|\Delta'(r_j)|$. Full proof in Appendix C; the multivariate coarea version is Corollary 1, proved in Appendix D. □

Why the boundary estimand matters substantively. β^{ov} is not a consolation prize; it is the effect in a policy-relevant population. The decision boundary $\{\Delta(H) = 0\}$ collects the histories at which the deployed policy is genuinely undecided, the algorithmic analogue of clinical equipoise (Li et al., 2018). Three consequences. First, *marginal policy evaluation*: any small change to the policy, a regulation, a guardrail, a reward-model update, a temperature change — alters behavior precisely on and near this boundary, so the effect among boundary histories is the first-order term in the value change of any local policy modification. Second, *auditability*: β^{ov} is the one turn-level causal quantity a regulator could estimate from logs alone with stable precision, which makes its scope and its limits worth stating exactly. Third, the political reading, stated as interpretation rather than result: histories far from the boundary, strongly reactant partisan contexts where the policy is most decisive, are exactly where population-level causal information vanishes; what deployment data can tell us about persuasion is confined to the persuadable margin as *the policy* defines it, not as the scientist would.

What Result B does *not* say. At fixed $\tau > 0$, an IPW estimator with true propensities targets the *population* effect as $n \to \infty$, the estimand does not silently become the boundary effect because the policy is sharp. Numerically: $\tau = 0.3$, $n = 4 \times 10^6$, $\beta(H) = 1 + |H|$ gives IPW $1.792 \approx$ ATE $1.798 \neq \beta^{\text{ov}} = 1.356$. Formal identification and practical estimability are different properties; our claims are about the *rate* of information loss and the *location* of surviving information, never about a change of target.

The joint limit is the real object. The failure surface lives in (n, τ): with $n \gg e^{\delta/\tau}$ the population effect is estimable (at exploding variance); near $n \sim e^{\delta/\tau}$ variance explodes; with

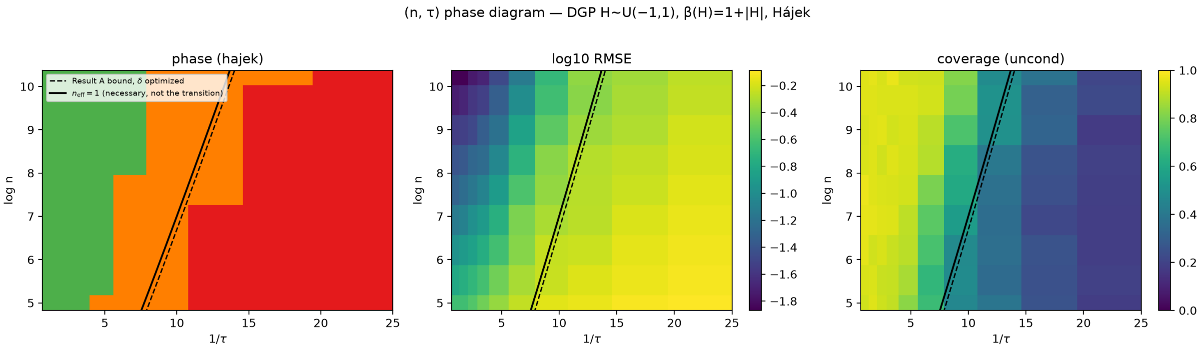

Figure 1: (n, τ) phase diagram; DGP $H \sim U(-1, 1)$, $\beta(H) = 1 + |H|$, Hájek estimator, 400 replications per cell. Left: regime classification (green estimable, orange variance explosion, red biased-stable). Middle, right: $\log_{10}$ RMSE and unconditional coverage. Two reference lines, neither of which is the observed transition. Dashed: the Proposition 2 bound with δ optimized, $\log n = 1/\tau - 1 - \log(1/\tau)$. Solid: the $n_{\text{eff}} = 1$ contour $\log \mathbb{E}[1/p_\tau] = \log(1 + \tau \sinh(1/\tau))$, a necessary condition only; measured onsets of estimability lie 3–5 nats above it. Regime thresholds (relative bias 0.10/0.20, coverage 0.85) are presentation choices defended by the RMSE and coverage panels.

$n \ll e^{\delta/\tau}$ rare crossover actions never appear in the sample and estimators become *biased while looking stable*, the deceptive regime. Figure 1 shows the three regimes for the main design $H \sim U(-1, 1)$, $\Delta(H) = H$, with two *reference* lines. The dashed line is the Proposition 2 bound with δ chosen optimally: since $\mathbb{P}(\mathcal{H}_\delta) = 1 - \delta$ here, $\max_\delta (1 - \delta) e^{\delta/\tau}$ is attained at $\delta = 1 - \tau$, giving $\log n = 1/\tau - 1 - \log(1/\tau)$. (Fixing $\delta = 1$, as an earlier version did, puts $\mathbb{P}(\mathcal{H}_\delta) = 0$ and makes the bound identically zero.) The solid line is the $n_{\text{eff}} = 1$ contour $\log \mathbb{E}[1/p_\tau] = \log(1 + \tau \sinh(1/\tau))$, exact for this design. We stress that neither line is the observed transition: the $n_{\text{eff}} = 1$ contour is a *necessary* condition, and on a 13×13 grid the measured onsets of estimability lie 3–5 nats above it. Which functional of the decisiveness distribution governs the transition is open (Section 8).

5 Do Deployed Policies Sit in This Regime?

The theory says that sufficiently decisive policies destroy the information needed to analyze them. Whether deployed systems are that decisive is an empirical question, and we can give a partial and carefully bounded answer.

Re-querying three RLHF-tuned open-weights models ($k = 20$ regenerations on each of 200 real debate contexts, 12,000 generations) and labelling each generation with an LLM judge, the per-context distribution over strategy labels is close to degenerate: median entropy zero and modal shares of 94.5–98.6% under our primary judge. Read literally, that would place these contexts far inside the collapse regime.

We do not read it literally, for three reasons that Appendix E develops. The raw generations are not degenerate at all, within a context no two are identical and mean pairwise lexical overlap is 0.19–0.26, so whatever collapses does so only after projection onto an eight-category label space. That projection is a measurement instrument with poor reliability ($\kappa = 0.13$ between two judges, and essentially no agreement once both-modal pairs are removed), and its modal category is near-tautological with the prompt used to generate the text. And at $k = 20$ a strategy with true probability 0.02 goes unobserved 67% of the time, so "never observed" does not license "very rare": the defensible one-sided statement from 0 of 20 is $\mathbb{E}[1/p] \geq 7.2$, not the 42 that the smoothed point estimate suggests.

What survives is weaker than the headline and still relevant: deployed generative policies are decisive enough, at the level of coarse behavioral categories, that minority-action contrasts would require sample sizes on the order of 10^4–10^5 conversations. That is a statement about where to look for trouble, not a measurement of any deployment's decisiveness, and it is the reason the design of Section 6 is worth the cost rather than a demonstration that it is necessary. An audit of 56,831 deployed persuasion conversations makes the companion point on the design side: what was randomized in those studies was an *instruction*, and it explains only 12–32% of the variance in the behavior actually realized.

6 Design Consequence: Generative Micro-Randomization

The obstruction is created by the deployed policy, so no estimator removes it. Each failure above does, however, correspond to something a design can supply. Uniform turn-level randomization restores both support and overlap, which answers Sections 3.1 and 3.2. Exogenous assignment of the turn's action removes sequential confounding. Regime-defined estimands resolve the ill-posedness of observational labels. And a fidelity protocol with assignment-level analysis handles the fact that the treatment is generated rather than administered.

We call the resulting design a generative micro-randomized trial. Its substance is four commitments. The generator is randomized at the conversation level, with at least one open-weights arm for reproducibility. The turn action is randomized uniformly, with assignment, probability, and realization all logged. The evidence available to the generator is a locked bank of verified fact units, and the sampling rule over that bank is specified as part of the treatment regime rather than left to the generator. Finally, each utterance passes a fidelity gate before it is sent: inspect, regenerate once on mismatch, send with a log entry on second failure.

The last commitment changes what is identified, and the change should be stated rather than absorbed. The gate is not quality control applied to a treatment; it is part of the treatment kernel. What an intention-to-treat analysis identifies is therefore the effect of assignment to a generate-and-check protocol, not the effect of a realized sentence. This is weaker than it first appears only if one expected the latter, which the argument of Section 2 says was never available.

Two design risks follow directly. Fidelity failures that differ across cells contaminate the contrast between them, so the pre-registration fixes a floor per cell and demotes contrasts that miss it. And a gate classifier whose errors correlate with assignment distorts the treatment it is supposed to protect, so gate decisions are logged and a no-gate sensitivity analysis is pre-specified. Naming the dose honestly is a third: manipulating the number of facts also moves length and cognitive load, so the manipulated quantity is a high-information communication package, not information as such.

Estimation follows the design rather than the logs. Proximal excursions are estimated by weighted centered least squares under the known randomization (Boruvka et al., 2018; Qian et al., 2022); distal outcomes call for distal excursion estimators (Qian, 2025) rather than terminal-outcome WCLS; policy values, if reported at all, are restricted to two or three pre-specified pairs with cross-fitted doubly robust estimation. Every guarantee here is relative to the design's known ρ. Nothing in this section rescues transcript-only analysis, which is the point.

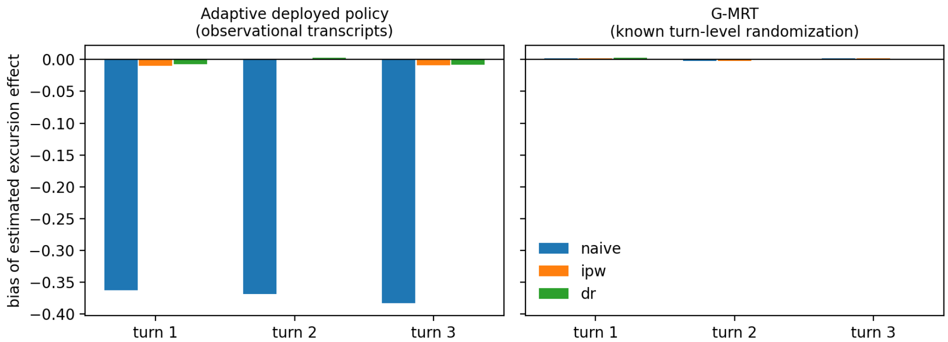

Figure 2: Sign reversal under an adaptive policy: naive turn-level contrasts versus known-ρ IPW/DR.

7 Simulation

Four executed studies; theory-to-DGP correspondence is declared object-by-object in the repository.

(i) Sign reversal under adaptivity. A defensiveness-adaptive policy makes the naive turn-level contrast wrong in *sign* (truth $+0.068$; naive -0.294, coverage 0.0) while known-ρ IPW/DR recover it (bias ≤ 0.01, coverage ≈ 0.95–0.97), the confounding face of the problem, and the G-MRT's licence (Figure 2).

(ii) Two-state temperature collapse. Along a τ grid calibrated so $\tau = 1$ reproduces the adaptive policy of (i): IPW SE grows exponentially, tracking a scaled inverse-propensity diagnostic curve (labeled as such, it is not the efficiency bound); the naive estimator becomes *more confident while more wrong* as τ falls (coverage $\to 0$ with shrinking SE) (Figure 3). Metric discipline: global arm-specific Kish ESS (theory object), minimum stratum-arm cell occupancy (rare-crossover diagnostic, not Kish), $\mathbb{E}[p(1-p)]$ (Proposition 5 object), and $\mathbb{E}[2\min(p, 1-p)]$ (diagnostic) are reported as four separate columns, coverage both conditional and unconditional on estimability, with zero-cell rates alongside. One finding we flag for practice: in the collapse regime the *realized* control-arm Kish ESS is non-monotone in τ: it falls to 0.028 at $\tau = 0.40$, while the zero-stratum-arm rate is still 0.00, and then *rebounds* to 0.037, 0.372, 0.534 at $\tau = 0.30, 0.22, 0.15$, over exactly the range where the zero-stratum-arm rate climbs $0.06 \to 0.70 \to 0.99$. Surviving weights homogenize as rare crossovers vanish from the sample, so a healthy-looking realized ESS co-occurring with high zero-cell rates is itself the signature of the deceptive regime. ESS should never be reported without zero-cell rates.

(iii) Continuous-boundary battery. ($H \sim U(-1, 1)$ main, $N(0, 1)$ robustness; $\beta(H) \in \{1,\, 1 + e^{-H^2},\, 1 + |H|\}$; 400 replications per cell.) At $n = 32{,}000$: all population-targeting estimators are unbiased with nominal coverage down to $\tau \approx 0.15$; at $\tau = 0.045$ Horvitz–Thompson has SD 2.95 (variance explosion) while Hájek shows bias 0.46 with SD 0.30, biased-yet-stable; oracle AIPW retains nominal coverage throughout but does not rescue *precision*, its SD inflating from 0.012 to 1.38 (a factor of 116) as τ falls — which is the behaviour Proposition 2 predicts, and confirms that the efficiency loss is not an IPW artifact; the oracle outcome-regression line is

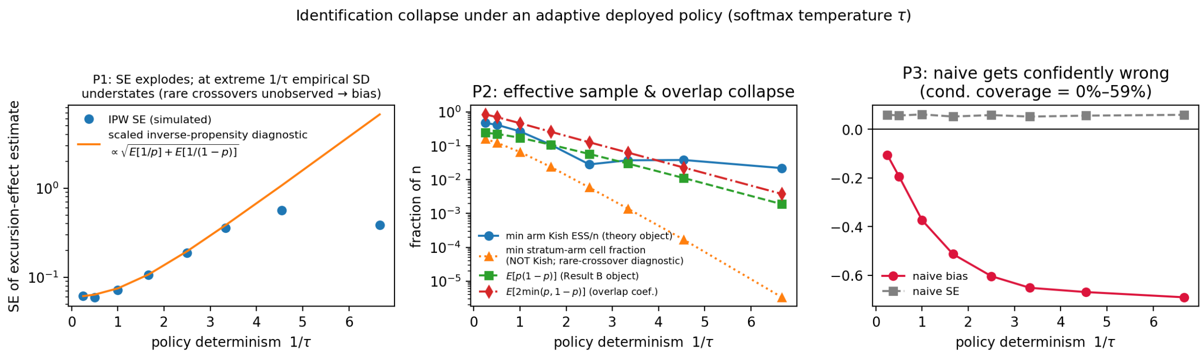

Figure 3: Two-state temperature collapse: predictions P1–P3.

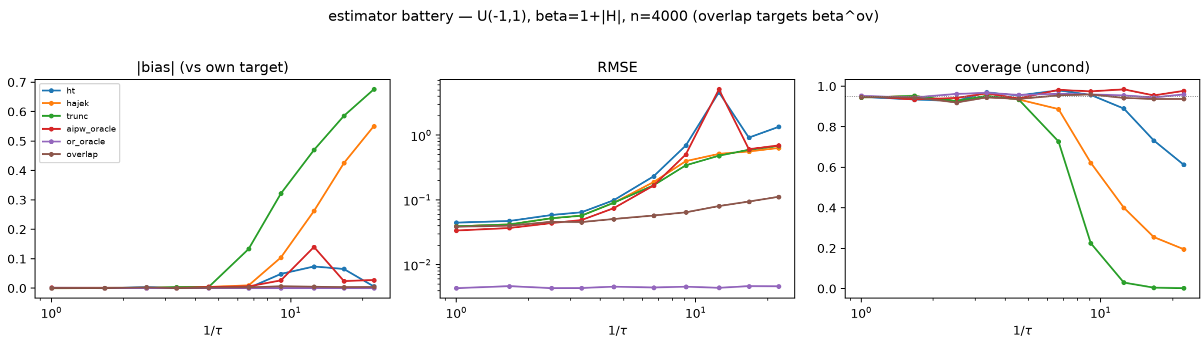

Figure 4: Estimator battery on the continuous-boundary design, $n = 4{,}000$ (bias against each estimator's own target; the overlap estimator targets β^{ov}). The numerical values quoted in the text are for $n = 32{,}000$, the largest cell of the grid.

flat by construction, the (A3) escape made visible, identification bought by assumption; and the overlap estimator remains unbiased with nominal coverage *for its own target* β^{ov} at every τ, with $\beta_\tau^{\text{ov}} \to \beta(0) = 1$ as Proposition 5 predicts (Figure 4). Under $N(0,1)$ the collapse is earlier and, at extreme τ, propensities underflow to machine 0/1 so IPW becomes numerically undefined, reported as estimability 0, not dropped.

(iv) Phase diagram. The three regimes separate cleanly over the (n, τ) grid (Figure 1). We report the reference lines without claiming either is the transition: measured onsets of estimability lie 3–5 nats above the $n_{\text{eff}} = 1$ contour, and on a widened 13×13 grid neither a mean functional $(\log \mathbb{E}[e^D])$ nor an upper-tail functional of D reproduces a stable offset across the $U(-1,1)$ and $N(0,1)$ designs. We therefore present the phase structure as a simulation finding and leave its asymptotic characterization open.

8 Discussion

The companion experiment. The companion paper implements the 2×2 G-MRT: pilot $n = 80$–120 (function verification: assignment mechanics, fidelity rates, bank leakage, dropout, not effect detection), demonstration $n = 180$–250 with pre-registered minority contrasts, confirmatory sample from post-pilot power simulation. Pre-registration locks the fidelity floor, blocking factors (issue), attrition handling (IPW-for-attrition), and reactivity checks (randomizing measurement itself in pilot). Ethics: persuasion toward *accuracy* on factual political questions with a fact-checked locked bank (cf. Costello et al., 2024); no deception arms; disclosure of AI interlocutor; IRB before pilot. The three Study 0 analyses involve only public data and no new human subjects.

What we have deliberately not done. Effects that transport across model versions would require treating the generator itself as a treatment arm, which cross-model heterogeneity suggests is a live question (Chen et al., 2026) but a different one. Adjustment on internal representations is the approach of Imai and Nakamura (2026) and Nakamura and Imai (2026), and inherits its own overlap problems (Tierney et al., 2025); our results say nothing against it, only that it solves a different failure. Simultaneous inference across a family of excursion contrasts, and policy learning under epistemic constraints, are both natural continuations we have kept out to preserve a single argument.

Limitations. *The impossibility is asymptotic in the policy, not the sample.* Proposition 2 bounds the information constant at a fixed policy, where the excursion effect remains regularly $\sqrt{n}$-estimable; Proposition 4 obtains a genuine impossibility only along sequences whose decisiveness grows with n. Whether a particular deployment is in that regime is an empirical question our Study 0 can motivate but not settle.

We bound impossibility, not achievability. Proposition 4 says when estimation must fail; it does not say when it becomes easy, and the observed onset of estimability in simulation is the latter. The two are separated by a design-dependent amount: the gap between the lower-bound functional M and the oracle-IPW variance factor $\mathbb{E}[e^D]$ is 1.00 nats for $U(-1, 1)$ and 2.06 for $N(0, 1)$. No single functional can therefore track both curves, and a matching upper bound is left open.

Proposition 3 rests on a recurrence condition in tension with our own framing. Condition (M1) requires histories to remain in the high-gap set under *every* action, i.e. that treatment does not move the history state much, which is close to the opposite of the premise in this paper's title. Its per-turn factor is $\rho^2_{\min} e^D p_\delta$, and p_δ is not estimated anywhere here; multi-turn compounding should be read as conditional on an unverified quantity.

Study 0 measures a projection, not a policy. The entropy and decisiveness figures in Section 5 are properties of an eight-category strategy label assigned by an LLM judge, not of the generation policy itself. Three specific threats: the modal category (rebuttal) is near-tautological with the generation prompt, which asks for a rebuttal; the two judges we ran agree at $\kappa = 0.13$ overall and not at all once both-modal pairs are excluded, and the headline entropy figures move materially between them; and at $k = 20$ regenerations a strategy with true probability 0.02 is unobserved with probability 0.67, so "never observed" is not "very rare." At the level of raw text the generations are not degenerate at all, within-context duplicates are absent and pairwise lexical overlap is low. Any collapse we report is a collapse *after* projection to the label space. The re-queried models are also open-weights stand-ins for the deployed system, not the deployed system. We therefore use Study 0 to motivate and locate, not to establish; human double-coding and higher-k re-query on the censored subset are the obvious next steps and are not yet done.

Study 0-c measures a bundled dose. Realized fact counts covary with AI verbosity ($r = 0.49$ and 0.71 in studies 2 and 3), so the within-assignment variance we report is not purely instruction–

behavior slippage; adjusting for message length *raises* η^2 rather than lowering it.

Data and code availability. All numbers reported here are produced by code in the project repository, which contains the simulation sources, the analysis pipeline, the re-query outputs and judge labels, and the derived result files. The DebateGPT reproduction uses the public release of Salvi et al. (2025); the deployment audit uses the public release of Hackenburg et al. (2025), and the 56,831 conversations are the released persuasion rows, not that paper's full sample. Simulation entry points and expected runtimes are documented in the repository README.

The result that organizes the rest is that adaptive systems produce the least population-level causal information exactly where they are most decisive, and that what survives is not a weaker estimate of the population effect but a boundary effect at a threshold the system chose. A deployment that has learned its job well is, for this purpose, a deployment that has learned to stop informing us about it. Turn-level causal knowledge about conversational AI therefore has to be randomized into the conversation rather than recovered from it.

A Efficient influence function and the decisiveness bound

Setup. Observe $O = (H, A, Y)$ with $A \in \mathcal{A}$, $|\mathcal{A}| = K < \infty$, and $\pi(a \mid h) = \mathbb{P}(A = a \mid H = h) > 0$ a.s. Write $\mu(a, h) = \mathbb{E}[Y \mid A = a, H = h]$ and $\sigma^2(a, h) = \mathrm{Var}(Y \mid A = a, H = h)$; the model is nonparametric. In the multi-turn setting read $H = H_t$, Y as the proximal outcome, and all statements on the availability stratum $\{I_t = 1\}$.

A regime is a known action distribution $q(\cdot \mid h)$ on $\mathcal{A}$ — a stochastic intervention — with value $\psi(q) = \mathbb{E}\big[\sum_a q(a \mid H)\, \mu(a, H)\big]$. The form estimand of Section 2 is $\beta^{\text{form}} = \psi(q_1) - \psi(q_0)$ with $q_1(f, d \mid h) = \mathbf{1}\{f = \text{assertive}\}\rho_D(d)$ and q_0 likewise for interrogative. Note that the dose is marginalized under the *reference* ρ_D, not under π; this is what makes the cell propensities, rather than the form-marginal, the relevant quantities.

Proposition 6 (Efficient influence function). *$\psi(q)$ is pathwise differentiable with efficient influence function*

$$\varphi_q(O) \;=\; \sum_a q(a \mid H)\mu(a, H) \;-\; \psi(q) \;+\; \frac{q(A \mid H)}{\pi(A \mid H)}\big(Y - \mu(A, H)\big),$$

so that for $\beta = \psi(q_1) - \psi(q_0)$ the semiparametric efficiency bound is

$$V(\beta) \;=\; \mathrm{Var}\big(m_1(H) - m_0(H)\big) \;+\; \mathbb{E}\Big[\textstyle\sum_a \big(q_1(a \mid H) - q_0(a \mid H)\big)^2 \frac{\sigma^2(a,H)}{\pi(a|H)}\Big], \tag{1}$$

where $m_j(h) = \sum_a q_j(a \mid h)\mu(a, h)$.

Proof. $\psi(q)$ depends on the observed-data law only through the marginal of H and the outcome regression μ. Decompose the tangent space as $\mathcal{T} = \mathcal{T}_H \oplus \mathcal{T}_{A|H} \oplus \mathcal{T}_{Y|A,H}$ and differentiate along a regular parametric submodel. The $\mathcal{T}_H$ component contributes $\sum_a q\mu - \psi$; the $\mathcal{T}_{Y|A,H}$ component contributes $(q/\pi)(Y - \mu)$. Because q is a known function and ψ does not depend on π, the $\mathcal{T}_{A|H}$ component vanishes — this holds even though q may depend on h. One checks $\mathbb{E}[\varphi_q] = 0$ and $\varphi_q \in \mathcal{T}$, so φ_q is the EIF. The two components are orthogonal, giving $\mathrm{Var}(\varphi_q) = \mathrm{Var}(m_q(H)) + \mathbb{E}[(q/\pi)^2\sigma^2]$; expanding the second term over a and using linearity in q for the contrast yields (1). □

When q_1 and q_0 have disjoint support — as they do for a form or dose contrast — $(q_1 - q_0)^2 = q_1^2 + q_0^2$ and the sum in (1) runs over all K cells.

Definition 1 (Policy decisiveness)**.** For $\mathcal{A}^* = \mathrm{supp}(q_1) \cup \mathrm{supp}(q_0)$ set $D(h) := -\log \min_{a \in \mathcal{A}^*} \pi(a \mid h)$.

D is a functional of the behaviour policy alone and is identified from logs; neither τ nor a softmax parameterization enters.

Proof of Proposition 2. Discard the first term of (1). On $\{D \geq d\}$, $\sum_a (q_1 - q_0)^2 \sigma^2/\pi \geq \underline{\sigma}^2 \rho_{\min}^2 \sum_{a \in \mathcal{A}^*} 1/\pi(a \mid H) \geq \underline{\sigma}^2 \rho_{\min}^2 / \min_{a \in \mathcal{A}^*} \pi(a \mid H) = \underline{\sigma}^2 \rho_{\min}^2 e^{D(H)} \geq \underline{\sigma}^2 \rho_{\min}^2 e^d$; multiply by $\mathbb{P}\{D \geq d\} \geq c$. $\square$

Remark 5 (Recovering Hahn (1998)). With $K = 2$, $q_1 = \delta_1$, $q_0 = \delta_0$ (so $\rho_{\min} = 1$), (1) is $\mathrm{Var}(\mu_1 - \mu_0) + \mathbb{E}[\sigma_1^2/\pi(1 \mid H) + \sigma_0^2/\pi(0 \mid H)]$, i.e. the ATE bound of Hahn (1998). The present paper therefore does not import that bound for a different estimand; it derives the bound for the excursion estimand and recovers Hahn's as a special case. If π is a softmax at temperature τ and $K = 2$, then $D = |\Delta|/\tau + O(e^{-|\Delta|/\tau})$ and the $e^{\delta/\tau}$ form is recovered; the converse fails, so Proposition 2 applies to sharply adaptive policies that are not softmax (truncated or constrained decoding, learned $\arg\max$ rules, hard filters). For $K = 2$ one also has the exact identity $1/\{p(1-p)\} = 2 + 2\cosh(\mathrm{logit}\, p)$.

Remark 6 (Scope of Proposition 2). Proposition 2 is a statement about the asymptotic variance at a *fixed* policy. Whenever $\pi(a \mid h) > 0$ everywhere, $V(\beta) < \infty$ and β remains regularly $\sqrt{n}$-estimable: on its own this is a constant factor, not an impossibility. The impossibility statement requires drifting sequences π_n and is Proposition 4 below; the closest antecedent for the irregular behaviour it describes is Khan and Tamer (2010).

Proof of Proposition 4. Fix $d > 0$, let $\mathcal{H}_d = \{D \geq d\}$ and $c = \mathbb{P}(\mathcal{H}_d) > 0$, and let a^* attain the minimum in Definition 1, so $\pi(a^* \mid h) \leq e^{-d}$ on $\mathcal{H}_d$. Define two laws P_0, P_1 that agree in the marginal of H, in π, and in $\mu(a,h)$ for every (a,h) except one: $\mu^{(0)}(a^*, h) = 0$ and $\mu^{(1)}(a^*, h) = \varepsilon$ for $h \in \mathcal{H}_d$.

Separation in the estimand. Since $\psi(q) = \mathbb{E}[\sum_a q(a \mid H)\mu(a, H)]$,

$$\beta(P_1) - \beta(P_0) \;=\; \mathbb{E}\big[q(a^* \mid H)\,\varepsilon\,\mathbf{1}\{H \in \mathcal{H}_d\}\big] \;\geq\; \rho_{\min}\, c\, \varepsilon \;=:\; \Delta.$$

Proximity in the data. The observed density factorizes as $P(H)\,\pi(A \mid H)\,N(Y; \mu_A(H), \sigma^2)$, and P_0, P_1 share the first two factors, so only the $(a^*, \mathcal{H}_d)$ cell contributes to the divergence:

$$\mathrm{KL}(P_0 \| P_1) = \mathbb{E}_H\Big[\textstyle\sum_a \pi(a \mid H)\,\mathrm{KL}\big(N(\mu_a^{(0)}, \sigma^2) \,\|\, N(\mu_a^{(1)}, \sigma^2)\big)\Big] = c\,\pi(a^* \mid h)\,\frac{\varepsilon^2}{2\sigma^2} \leq \frac{c\,e^{-d}\varepsilon^2}{2\sigma^2}.$$

By tensorization $\mathrm{KL}(P_0^n \| P_1^n) = n\,\mathrm{KL}(P_0 \| P_1)$. Choose $\varepsilon = \sigma\sqrt{e^d/(nc)}$, so that this equals $1/2$; Pinsker's inequality then gives $\mathrm{TV}(P_0^n, P_1^n) \leq \sqrt{\mathrm{KL}/2} = 1/2$.

Le Cam. The two-point bound $\inf_{\hat\beta} \sup_P \mathbb{E}|\hat\beta - \beta| \geq (\Delta/2)(1 - \mathrm{TV}) \geq \Delta/4$ with $\Delta = \rho_{\min} c \varepsilon = \rho_{\min}\sigma\sqrt{c\,e^d/n}$. Optimizing over d replaces $c\,e^d$ by $M(\pi)$. The constraint $|\mu| \leq B$ caps ε at B, giving the stated minimum (for the two-point method see Tsybakov, 2009, §2.2). $\square$ $\square$

B Support failure (Proposition 1)

Take $T = 1$, $H \in \{h_0, h_1\}$ with $\mathbb{P}(H = h_1) > 0$, $\pi^{\text{dep}}(1 \mid h_0) = 1/2$ and $\pi^{\text{dep}}(1 \mid h_1) = 0$, and $Y(a) \mid H = h \sim N(\mu_a(h), 1)$. Let two laws P_1, P_2 agree in every component except $\mu_1(h_1)$, equal to c_1 under P_1 and $c_2 \neq c_1$ under P_2. The observed-data density factorizes as $P(H)\,\pi^{\text{dep}}(A \mid H)\,N(y; \mu_A(H), 1)$ and the stratum $\{A = 1, H = h_1\}$ has probability zero, so $\mu_1(h_1)$ does not appear: P_1 and P_2 induce identical observed-data laws. Their uniform-reference excursion effects differ by $\mathbb{P}(h_1)(c_2 - c_1) \neq 0$. Hence no functional of the observed-data law equals the excursion effect, which is the claim. No latent confounder is used: the failure is support, not confounding. The multi-turn version replaces H by $(X, R_{1:t-1})$ and recovers the standard time-varying-confounding form of Robins (1986); the contribution here is not the structure but the claim that the triggering condition is structurally frequent in RLHF-tuned deployment, which is an empirical claim (Section 5) and not a theorem.

C Localization in one dimension (Proposition 5)

Throughout $\omega_\tau(h) = p_\tau(1 - p_\tau) = \frac{1}{4}\text{sech}^2\big(\frac{\Delta(h)}{2\tau}\big) = \frac{1}{\tau}K\big(\frac{\Delta(h)}{\tau}\big)$ with $K(u) = 1/[4\cosh^2(u/2)]$, an exact identity. Two elementary facts: $\int_{\mathbb{R}} K = [\frac{1}{2}\tanh(u/2)]_{-\infty}^{\infty} = 1$, and $\cosh^2(u/2) \geq e^{|u|}/4$ gives $K(u) \leq e^{-|u|}$.

Fix a bounded continuous g and put $I_\tau(g) = \int \omega_\tau g f$. Split at $U_\epsilon = \{\text{dist}(h, \mathcal{R}) < \epsilon\}$. Outside U_ϵ, (B2$'$) gives $|\Delta| \geq \eta(\epsilon) > 0$, so $\omega_\tau \leq e^{-\eta/\tau}$ and, f and g being bounded, that contribution is $O(e^{-\eta/\tau}) = o(\tau)$. Inside, work near a single root r with $\Delta'(r) \neq 0$ (B2); by the inverse function theorem Δ is a C^1 diffeomorphism of a neighbourhood of r onto a neighbourhood of 0, so substituting $v = \Delta(h)$ and then $v = \tau u$,

$$\int_{U_\epsilon} \omega_\tau g f \, dh = \int K(u)\, \frac{(gf)(\Delta^{-1}(\tau u))}{|\Delta'(\Delta^{-1}(\tau u))|}\, \tau \, du.$$

The integrand is dominated by $K(u)\sup_{|h-r|\leq\epsilon} |gf/\Delta'|$, finite because $|\Delta'|$ is continuous and bounded away from 0 near r; by dominated convergence and continuity of f, g at r the integral tends to $g(r)f(r)/|\Delta'(r)|$. Hence $I_\tau(g) = \tau\, g(r)f(r)/|\Delta'(r)| + o(\tau)$. Summing over the finitely many roots (B2) gives $I_\tau(g) = \tau \sum_j g(r_j)f(r_j)/|\Delta'(r_j)| + o(\tau)$. Taking $g \equiv 1$ yields the statement for $\mathbb{E}[\omega_\tau]$, where (B4) $f(r) > 0$ for some root is exactly what prevents the leading term from vanishing and the expansion from degenerating to $0 = o(\tau)$. Ratios $I_\tau(g)/I_\tau(1)$ give weak convergence to $\sum_j w_j \delta_{r_j}$ with $w_j \propto f(r_j)/|\Delta'(r_j)|$, and $g = c$ (bounded, continuous near the roots, by (B3)) gives the limit of β_τ^{ov}. With a unique root at 0 this is a point mass and $\beta_\tau^{\text{ov}} \to c(0)$. $\square$

D Localization on a decision manifold (Corollary 1)

Let $H \in \mathbb{R}^d$ and $\mathcal{B} = \Delta^{-1}(0)$. By (C2), $\nabla\Delta \neq 0$ on $\mathcal{B}$, so 0 is a regular value and $\mathcal{B}$ is a C^1 hypersurface of dimension $d - 1$. On $U_{\epsilon_0} = \{|\Delta| < \epsilon_0\}$ the coarea formula (e.g. Evans and Gariepy, Thm. 3.4.2) gives, for integrable Φ,

$$\int_{U_{\epsilon_0}} \Phi \, dh = \int_{\mathbb{R}} \Big[\int_{\Delta^{-1}(s)} \frac{\Phi}{\|\nabla\Delta\|} \, d\mathcal{H}^{d-1} \Big] ds.$$

Apply this with $\Phi = \omega_\tau g f$. Since ω_τ depends on h only through $\Delta(h)$, it is constant on level sets and factors out: with $G(s) := \int_{\Delta^{-1}(s)} gf/\|\nabla\Delta\| \, d\mathcal{H}^{d-1}$,

$$\int_{U_{\epsilon_0}} \omega_\tau g f \, dh = \int_{-\epsilon_0}^{\epsilon_0} \tfrac{1}{\tau} K(s/\tau)\, G(s)\, ds = \int_{|u|\le \epsilon_0/\tau} K(u)\, G(\tau u)\, du \cdot \tau.$$

By (C1)–(C2) and the implicit function theorem the level sets $\Delta^{-1}(s)$ vary continuously in s near 0 along the normal flow of $\mathcal{B}$, and f, g, $\|\nabla\Delta\|$ are continuous with $\|\nabla\Delta\|$ bounded away from 0; hence G is continuous at 0 with $G(0) = \int_{\mathcal{B}} gf/\|\nabla\Delta\| \, d\mathcal{H}^{d-1}$. Since $\int K = 1$ and $K(u) \le e^{-|u|}$, dominated convergence gives $\int_{U_{\epsilon_0}} \omega_\tau g f = \tau\, G(0)\,(1 + o(1))$. The region outside U_{ϵ_0} contributes $O(e^{-\eta/\tau})$ by (C4) as in Appendix C. Taking $g \equiv 1$ gives the stated expansion of $\mathbb{E}[\omega_\tau]$; (C5) guarantees the limit is in $(0, \infty)$, so ratios converge and the normalized overlap measure tends weakly to the measure on $\mathcal{B}$ with $\mathcal{H}^{d-1}$-density proportional to $f/\|\nabla\Delta\|$. Taking $g = c$ gives the limit of β_τ^{ov}. For $d = 1$, $\mathcal{H}^0$ is counting measure and $\|\nabla\Delta\| = |\Delta'|$, recovering Appendix C. □

E Study 0 in full

This appendix reports the three descriptive analyses summarized in Section 5, together with the reasons we treat them as motivating rather than confirmatory.

Three empirical bridges locate deployed systems on the diagram. Their epistemic status is *motivating*: they estimate proxies of policy decisiveness and instruction–behavior slippage, not the parameters of Sections 3.1–4.

(0-a) Reproduction. Our pipeline reproduces the direction of Salvi et al. (2025) on the released data ($n = 750$ rows from 600 debates): the personalized Human-AI arm shows the largest shift toward the opponent's side. Because the Human-Human arms contribute two rows per debate, we report cluster-robust rather than crude inference. On the binarized outcome $\mathbf{1}\{\Delta > 0\}$ with sandwich standard errors clustered on debate, the personalized Human-AI odds ratio against Human-Human is 2.18 (95% CI $[1.38, 3.43]$, $p = 0.0008$); the unpersonalized AI arm is 1.45 ($[0.90, 2.35]$) and personalized Human-Human is 0.95 ($[0.57, 1.61]$). Binarizing discards the 49.5% of rows with no change and all magnitudes; on the full -4 to $+4$ scale a proportional-odds model gives a smaller estimate for the same contrast, 1.68 ($[1.16, 2.42]$, $p = 0.006$), which is the number we would cite if one were needed. The direction survives both specifications; the magnitude is specification-dependent, and the dichotomized version is the flattering one.

(0-b) Re-query decisiveness. For 200 rebuttal contexts stratified by treatment and prior-agreement strength, we regenerate $k = 20$ responses per context and classify the persuasion strategy of each. Across three RLHF-tuned open-weights families the per-context strategy distribution is near-degenerate (Table 1).

Substituting per-strategy $\hat{p}_s$ into the effective-sample formula (Dirichlet-smoothed, $k = 20$ resolution): the modal strategy needs $\sim 3 \times 10^3$ conversations per contrast arm and minority strategies are on the order of 10^5 (7.7×10^4 to 1.2×10^5 across strategies and judges), with a two-minority contrast needing the *sum* of two such terms. Three caveats are essential. First, these are illustrative insertions into Proposition 2's formula, not estimates of any deployment's policy. Second, the magnitude is set by choices we make, not by the data: $n_{\mathrm{req}} = \sigma^2\, \mathbb{E}[1/\hat{p}]/\mathrm{SE}^{*2}$ with $\sigma^2 = 1$ and $\mathrm{SE}^* = 0.02$, and for a strategy never observed in k draws the smoothed $\hat{p}$

model	median entropy (bits)	zero-entropy contexts	modal share
Qwen2.5-7B-Instruct	0.0	89.5%	REBT 98.6%
Mistral-7B-Instruct-v0.3	0.0	58.0%	REBT 94.5%
Phi-3.5-mini-instruct	0.0	75.5%	REBT 97.3%

Table 1: Re-query decisiveness across model families: 200 contexts $\times$ $k = 20$ regenerations each; strategies classified by a single 32B judge. Low entropy recurs across families — the observable signature of "structurally frequent."

is exactly $\alpha/(k + \alpha K)$, so $\mathbb{E}[1/\hat{p}]$ is the constant $(k + \alpha K)/\alpha = 48$ regardless of how rare the strategy truly is; halving α to 0.1 moves the figure to $\sim 5 \times 10^5$. Third, for the same reason these numbers say "below the resolution of $k = 20$," not "this rare."

Placing contexts on the phase diagram. Only the composite Δ/τ — the score gap in temperature units, exactly the phase diagram's abscissa — is identified from choice frequencies; τ alone is not. Estimating the two-action reduction $\widehat{\Delta/\tau} = \log[(c_1 + \alpha)/(c_2 + \alpha)]$ per context yields a median of 3.71 for all three models — which is the *resolution ceiling* of $k = 20$ draws: 58–89.5% of contexts are right-censored there (the runner-up action was never realized). The reflexive reading is that measuring a deployed policy's decisiveness runs into the same information limit that blocks effect estimation: with the runner-up never realized, k draws cannot distinguish "rare" from "absent." We are deliberately careful about what that licenses. The value $3.71 = \log\{(k + \alpha)/\alpha\}$ is the arithmetic ceiling of the estimator at $k = 20$, not a property of any model, which is why all three models return it. A defensible one-sided statement uses the exact binomial instead: observing the runner-up 0 times in 20 draws gives a Clopper–Pearson 95% upper bound $p_2 \leq 0.139$, hence $\Delta/\tau \geq 1.82$ and $\mathbb{E}[1/p] \geq 7.2$ — an order of magnitude below the ceiling value. We therefore do not propagate the ceiling figure through Proposition 3; with $K = 8$ strategy categories that proposition's compounding condition $\Delta/\tau > \log(K^2/p_\delta)$ is not met by the measured values in any case.

Two robustness results discipline these numbers. *Judge sensitivity*: relabeling all 4,000 Qwen responses with a 32B judge yields raw agreement 0.903 but $\kappa = 0.13$, and $\kappa \approx 0$ once both-modal pairs are excluded — judges agree on the dominant label and disagree almost completely on minority-label identity. Aggregate quantities are nevertheless stable (per-strategy n_{req} ratios 0.88–1.52 across judges). We therefore report entropy, modal share, and n_{req} — and we do not interpret minority-label identity before human double-coding (pre-registered $\kappa \geq 0.6$ on 200 turns). *Label-error simulation*: propagating misclassification rates $q \in \{0.05, 0.1, 0.2\}$ through the pipeline (symmetric and majority-absorbing noise models anchored to the observed confusion pattern) moves rare-strategy n_{req} by -25% to $+6\%$ even at $q = 0.2$; symmetric noise biases toward optimism, so the 10^5 scale is conservative.

(0-c) Instruction $\neq$ behavior. In the released data of Hackenburg et al. (2025) (56,831 persuasion conversations across three studies), the randomized rhetoric instruction explains $\eta^2 = 0.20/0.12/0.32$ of realized fact-count variance in studies 1/2/3 — 68–88% of realized-dose variation occurs *within* assignment. Assignments also bundle: "information" realizes 8.3–21.8 facts on average while "moral reframing" realizes 1.6–2.2, and 52–56% of moral-reframing conversations realize zero facts. Where compliance flags exist, valence compliance is 72.7% (study 1) and 89.6% (study 3); grammar and on-topic compliance are above 92% in both. What was randomized was an instruction; the treatment that reached the participant was a generated behavior, substantially decoupled from it. This is the observational counterpart of the fidelity problem the G-MRT protocol addresses by design.

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